FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2024
FULL TEST – XI
PAPER –1
TEST DATE: 19-05-2024

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 51 questions.
- Each subject (PCM) has 17 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-B**.

Section – A (01 –03, 18 – 20, 35 – 37): *This section contains **NINE (9)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).*

Section – A (04 – 07, 21 – 24, 38 – 41): *This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.*

Section – A (08 – 11, 25 – 28, 42 – 45): *This section contains **TWELVE (12)** Matching List Type Questions. Each question has **FOUR** statements in **List-I** entries (P), (Q), (R) and (S) and **FIVE** statements in **List-II** entries (1), (2), (3), (4) and (5). The codes for lists have choices (A), (B), (C), (D) out of which, **ONLY ONE** of these four options is correct answer.*

Section – B (12 – 17, 29 – 34, 46 – 51): *This section contains **EIGHTEEN (18)** numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.*

MARKING SCHEME

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	If three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

Physics

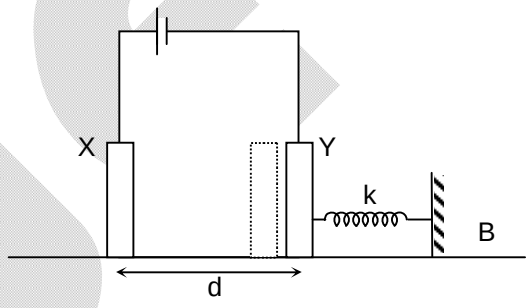
PART – I

SECTION – A
(One or More than one correct type)

*This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

1. In an experiment to determine the acceleration due to gravity 'g' the formula used for time period of a periodic motion is $T = 2\pi\sqrt{\frac{7(R-r)}{5g}}$. The values of R and r are measured to be (60 ± 1) mm and (10 ± 1) mm respectively. In five successive measurements, the time period is found to be 0.50 sec, 0.52 sec, 0.56 sec, 0.54 sec and 0.59 sec. The least count of the watch used for measurement time period is 0.01 sec. Which of the following statements is/are correct?
(A) error in measurement of r is 10%
(B) error in measurement T is approximately 5.45%
(C) error in measurement T is 2%
(D) error in the determined value of g is approximately 15%

2. Plate X of a capacitor is fixed and other plate is attached to an ideal spring of force constant 'k' whose other end is attached to a fixed support as shown in figure. Initially a battery of emf v_0 is connected to the plates of capacitor and the spring and capacitor plates in steady equilibrium position with separation between the plates is d. From this position is plate y is displaced slightly as shown and released, then choose the correct option(s). (Take 'A' area of plate and mass of plate as 'm')



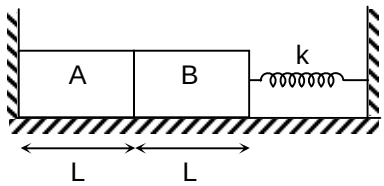
(A) In equilibrium the spring will have an extension of $\frac{A\varepsilon_0 v_0^2}{2kd^2}$

(B) Heat is produced in the circuit in subsequent motion.

(C) Plate attached with spring would oscillate in SHM with time period $2\pi\sqrt{\frac{Md^3}{2kd^3 - A\varepsilon_0 v_0^2}}$

(D) Plate attached with spring would oscillate in SHM with time period $2\pi\sqrt{\frac{Md^3}{kd^3 - A\varepsilon_0 v_0^2}}$

3. Two rods A and B having same length L and cross-sectional area A but different linear expansion coefficient α_A and α_B respectively are joined with an ideal spring as shown in figure. Now temperature of system is increased by an amount T_0. Initially, the spring is in its natural length and young's modulus of A and B are Y_A and Y_B respectively. Then



(A) Thermal stress developed in rod A and B is equal
(B) Thermal stress developed in rod A and B is unequal
(C) Compression in the spring will be $\frac{LT_0(\alpha_A + \alpha_B)}{\left[1 + \frac{kL}{A}\left(\frac{1}{Y_A} + \frac{1}{Y_B}\right)\right]}$
(D) Compression in the spring will be $\frac{2LT_0(\alpha_A + \alpha_B)}{\left[1 + \frac{kL}{A}\left(\frac{1}{Y_A} + \frac{1}{Y_B}\right)\right]}$

SECTION – A
(One Options Correct Type)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

4. A helicopter take off along the vertical with an acceleration a = 3 m/s^2 and zero initial velocity at time t = 0. At certain time t_0, pilot switches off the engine. At the point of take off, the sound dies away at time t_1 = 30 sec. Speed of sound is 320 m/s. Choose the INCORRECT option.

(A) Pilot switches off the engine at height h = $\frac{3200}{3}$ m

(B) Pilot switches off the engine in time $t_0 = \frac{80}{3}$sec

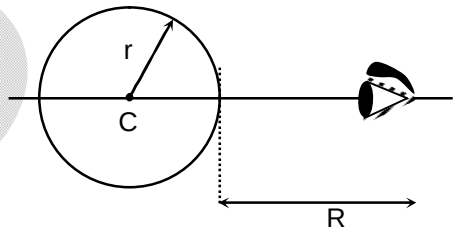
(C) Pilot switches off the engine at height h = $\frac{4000}{3}$ m

(D) Speed of helicopter at the moment when pilot switches off the engine is 80 m/s

5. In a resonance tube experiment, a 1 m long pipe is resonated with a tuning fork of frequency 340 Hz. Initially the pipe is completely filled with in water. A small hole is drilled at the bottom and water is allowed to leak. Is the radii of the pipe and hole is 2 cm and 1 mm respectively, then the time interval between the accuracne of first two resonance is (neglect end correction and take speed of sound 340 m/s, g = 10 m/s^2, $\sqrt{5} = 2.2$, $\sqrt{3} = 1.73$)

(A) 66 sec (B) 55 sec
(C) 46 sec (D) 75 sec

6. The figure shows the equatorial circle of a glass sphere of radius R having centre at C. The eye of observer in the plane of the circle at a distance R form surface. A small insect in crawling along equatorial circle. Refractive index of the glass is $\mu = \sqrt{2}$. The length (L) of the arc on the circle where insect lies so that its image is visible to the observer is



(A) $\frac{\pi R}{3}$ (B) $\frac{\pi R}{4}$
(C) $\frac{\pi R}{6}$ (D) $\frac{2\pi R}{3}$

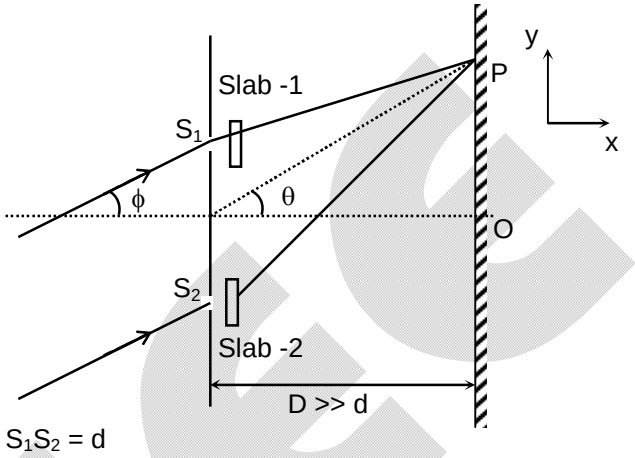
7. The diameter of a cylinder in measured using a vernier calipers with no zero error. It is found that the zero of the vernier scale lies between 5.10 and 5.15 cm of the main scale. The vernier scale has 50 divisions equivalent to 2.45 cm. The 24th division of the vernier scale exactly coincides with one of the main scale divisions. The diameter of the cylinder is

(A) 5.112 cm (B) 5.124 cm
(C) 5.136 cm (D) 5.148 cm

SECTION – A
(Matching List Type)

*This section contains **FOUR (04)** Matching List Type Questions. Each question has FOUR statements in **List-I** entries (P), (Q), (R) and (S) and FIVE statements in **List-II** entries (1), (2), (3), (4) and (5). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.*

8. Young's double slit experimental arrangement is shown in the diagram 'θ' represents the angular position of a point P on the screen. Also, note that two slabs made up of material of refractive indices n_1 and n_2 and each of thickness t are placed infront of the slits. Match the statements in List-I with the statements in List-II.

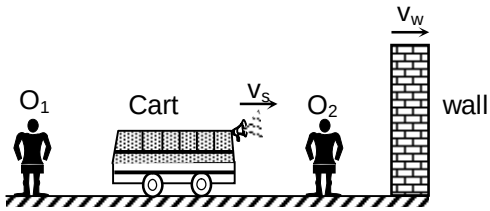


List –I		List –II	
(P)	Central maxima must be above point O	(1)	If $\phi = 30°$ and $n_1 = n_2$
(Q)	Central maxima must be below point O	(2)	If $\phi = 30°$, $n_1 = 1.1$ and $n_2 = 1.6$
(R)	Central maxima must be at origin O	(3)	If $\phi = 30°$ and $n_1 = 1.3$, $n_2 = 1.5$
(S)	Central maxima lie at $y = D\left(1-\frac{0.8t}{d}\right)$	(4)	If $\phi = 0$ and $n_1 = n_2$
		(5)	If $\phi = 0$ and $n_1 < n_2$

The correct option is:

(A) (P) → (1) (Q) → (5) (R) → (4) (S) → (2)
(B) (P) → (3) (Q) → (5) (R) → (2) (S) → (4)
(C) (P) → (1) (Q) → (2) (R) → (4) (S) → (3)
(D) (P) → (4) (Q) → (5) (R) → (1) (S) → (2)

9. Observer O_1 and O_2 are at rest and the wall is moving with velocity v_W. Cart is moving with constant velocity v_S towards wall. The source of sound is in cart, the original frequency of the wave is f_0. Sound has velocity C with respect to ground (medium is stationary). Match the statements in List-I with the statements in List-II.



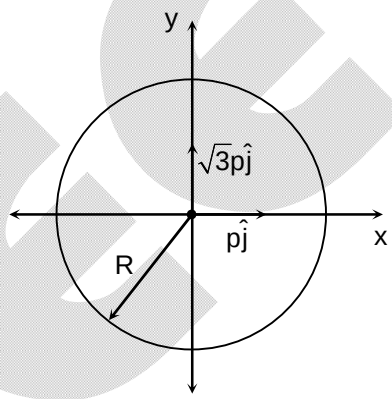
List –I		List –II	
(P)	Wavelength received by O_1 directly from cart	(1)	$\left(\frac{C-v_W}{C+v_W}\right)\left(\frac{C-v_S}{f}\right)$
(Q)	Wavelength received by O_2 directly from cart	(2)	$\left(\frac{C+v_S}{f}\right)$
(R)	Wavelength received by driver of the cart after reflection from wall	(3)	$\left(\frac{C+v_W}{C-v_W}\right)\left(\frac{C-v_S}{f}\right)$

(S)	Wavelength received by O_1 after reflection from wall	(4)	$\left(\frac{C-v_S}{f}\right)$
		(5)	$\left(\frac{C-v_W}{f}\right)$

The correct option is:

(A) (P) → (3) (Q) → (4) (R) → (2) (S) → (1)
(B) (P) → (3) (Q) → (2) (R) → (5) (S) → (4)
(C) (P) → (2) (Q) → (4) (R) → (3) (S) → (3)
(D) (P) → (2) (Q) → (5) (R) → (1) (S) → (5)

10. List-I gives a situation in which two dipoles of dipole moment $p\hat{i}$ and $\sqrt{3}p\hat{j}$ are placed at origin. A circle of radius R with centre at origin is drawn as shown in the figure. List-II gives coordinates of certain positions on the circle. Match the statements in List-I with the statements in List-II.



List –I		List –II	
(P)	The coordinate(s) of point on circle where potential is maximum	(1)	$\left(\frac{R}{2},\frac{\sqrt{3}R}{2}\right)$
(Q)	The coordinate(s) of point on circle where potential is zero	(2)	$\left(-\frac{R}{2},-\frac{R}{2}\right)$
(R)	The coordinate(s) of point on circle where magnitude of electric field intensity is $\frac{1}{4\pi\varepsilon_0}\frac{4p}{R^3}$	(3)	$\left(-\frac{\sqrt{3}R}{2},\frac{R}{2}\right)$
(S)	The coordinate(s) of point on circle where magnitude of electric field intensity is $\frac{1}{4\pi\varepsilon_0}\frac{2p}{R^3}$	(4)	$\left(\frac{R}{2},-\frac{R}{2}\right)$
		(5)	$\left(-\frac{\sqrt{3}R}{2},-\frac{R}{2}\right)$

The correct option is:

(A) (P) → (3) (Q) → (1) (R) → (4) (S) → (2)
(B) (P) → (1) (Q) → (2) (R) → (1) (S) → (3)
(C) (P) → (4) (Q) → (1) (R) → (4) (S) → (5)
(D) (P) → (1) (Q) → (3) (R) → (1) (S) → (3)

11. When photons of energy 4.25 eV strike the surface of a metal A, the ejected photoelectrons have maximum kinetic energy T_A eV and de-Broglie wavelength λ_A. The maximum kinetic energy of photoelectrons librated from another metal B by photons of energy 4.70 eV is T_B and $(T_B - T_A) = 1.50$ eV. If the de-Broglie wavelength of these photoelectrons is $\lambda_B = 2\lambda_A$. Work function of metal A is ϕ_A and ϕ_B for metal B. Match the statements in List-I with the statements in List-II.

List –I		List –II	
(P)	The value of T_A is	(1)	2.5 eV
(Q)	The value of T_B is	(2)	4.20 eV
(R)	The value of ϕ_A is	(3)	2.25 eV
(S)	The value of ϕ_B is	(4)	0.5 eV
		(5)	2 eV

The correct option is:

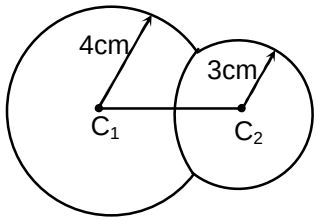
(A) (P) $\rightarrow$ (3) (Q) $\rightarrow$ (4) (R) $\rightarrow$ (2) (S) $\rightarrow$ (1)
(B) (P) $\rightarrow$ (3) (Q) $\rightarrow$ (1) (R) $\rightarrow$ (5) (S) $\rightarrow$ (4)
(C) (P) $\rightarrow$ (5) (Q) $\rightarrow$ (4) (R) $\rightarrow$ (3) (S) $\rightarrow$ (2)
(D) (P) $\rightarrow$ (5) (Q) $\rightarrow$ (1) (R) $\rightarrow$ (3) (S) $\rightarrow$ (2)

SECTION – B

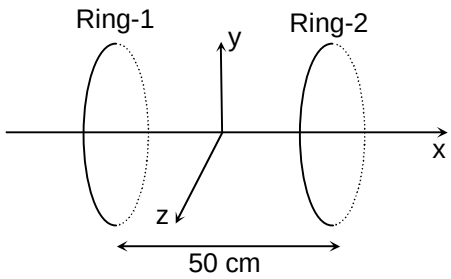
(Numerical Answer Type)

*This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.*

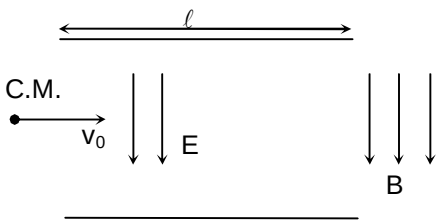
12. Two soap bubbles of radii 4 cm and 3 cm respectively are merged as shown in the figure and they are in equilibrium. If separation between the centres of both bubbles is $\sqrt{x}$. Find the value of x.



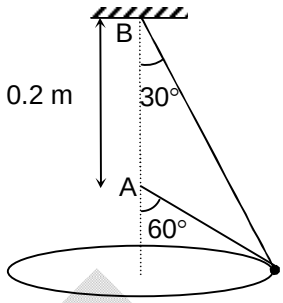
13. Two co-axial rings of same radius R = 10 cm placed parallel to y-z plane, such that x-axis in the axis of rings. Ring-1 carries current of 2 ampere and ring-2 carries a current 1 ampere as shown in the figure. The separation between the rings is d = 50 cm. Find the magnitude of $\int_{-\infty}^{+\infty} \frac{\vec{B}\cdot d\vec{x}}{\mu_0}$, where $\vec{B}$ is the net magnetic field at any point on the axis. (current in both the rings are in opposite direction)



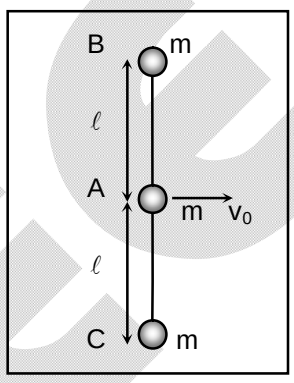
14. An electron accelerated by a potential difference v = 3.6 volt first enters a region of uniform electric field of a parallel plate. Capacitor whose plates extend over a length ℓ = 6 cm in the direction of initial velocity. The electric field is normal to the direction of initial velocity as shown in the figure and its strength varies with time E = $\alpha \times t$ (α = 3200 $Vm^{-1}s^{-1}$). Then the electron enters a region of uniform magnetic field of $B = \pi \times 10^{-9} T$. Find the pitch of motion of electron in magnetic field. Direction of magnetic field is same as that of electric field ($M_e = 9 \times 10^{-31}$ kg, $e = 1.6 \times 10^{-19}$C) (neglect the effect of induced magnetic field)



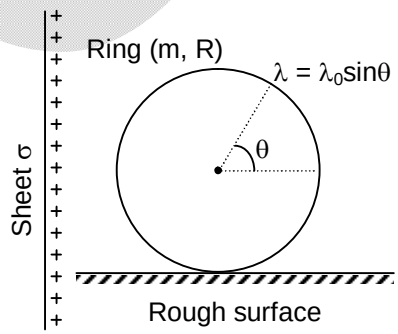
15. Two wires AC and BC are each tied to a particle at C and AB = 0.2m. The particle is made to revolve in a horizontal circle at a constant speed v. If both the wires are taut, the minimum value of v in m/s is ($g = 10\ m/s^2$)



16. Three identical ball having mass 'M' are tied by string are placed on smooth horizontal table. Impulse is applied on Ball A, that give speed v_0 to ball A as shown. Find the radius of curvature of ball B in ground frame just before collision between B and C. ($\ell = 2\sqrt{3}\,m$)



17. A non-conducting ring (mass m and radius R) having charge density $\lambda = \lambda_0 \sin\theta$, where λ_0 is a positive constant, is placed in front of an infinitely large uniformly charged non-conducting sheet (charge density σ). If the ring is in pure rolling motion, the acceleration of centre of mass of the ring is $\frac{x\lambda_0\sigma\pi R}{8m\varepsilon_0}$. Find the value of x.



Chemistry

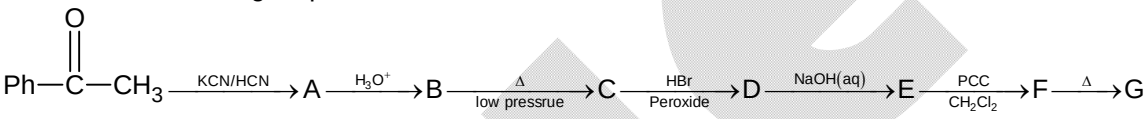
PART – II

SECTION – A
(One or More than one correct type)

*This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

18. Potassium cyanide is used for extraction of gold in the presence of air (O_2). Native gold in a solution containing cyanide ion (CN^-) forms $[Au(CN)_2]^-$. Which of the following option, is/are correct for extraction of gold from 0.1 kg of rock sample having 1.97% gold (w/w). [Molecular weight of Au = 197, K = 39, C = 12, N = 14, O = 16]
(A) 1.3 g KCN is required
(B) 2.6 g KCN is required
(C) 2.5×10^{-3} moles of O_2 is required
(D) 2×10^{-3} moles of KCN is required

19. Consider the following sequence of reaction



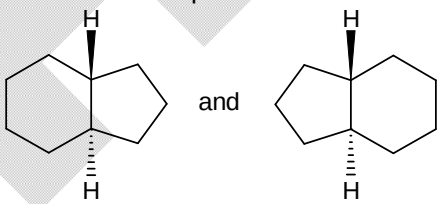
Choose the correct option(s):
(A) Mass % of H in compound G is 6.66
(B) Compound G will give positive Fehling test
(C) Both compound C and G will de-colourise $KMnO_4$
(D) Degree of unsaturation in compound F is 5

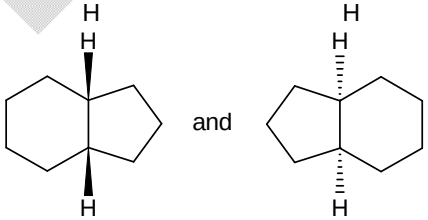
20. When a 4 ampere current is passed for 4 hrs between Ni electrodes in 1 litre of 1 M aq. solution of $Ni(NO_3)_2$
(A) There is no change in mass of either electrode after electrolysis
(B) There is no change in concentration of Ni^{2+} in electrolytic solution
(C) Concentration of Ni^{2+} in electrolytic solution decreases after electrolysis
(D) Mass of Ni-electrode (working as cathode) increases after electrolysis

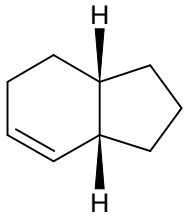
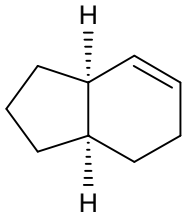
SECTION – A
(One Options Correct Type)

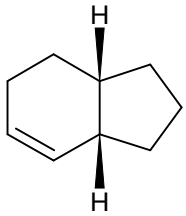
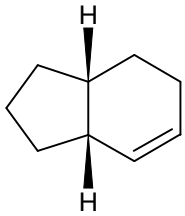
*This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

21. Choose the correct option

I.  and are enantiomers

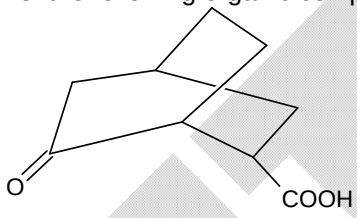
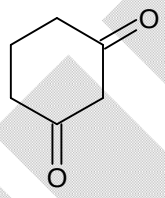
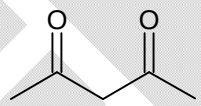
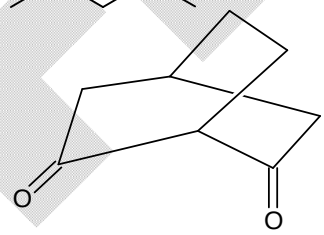
II.  and are enantiomers

III.  and  are enantiomers

IV.  and  are identical

(A) I, II and III only
(B) I and III only
(C) I, II and IV only
(D) II and IV only

22. Which of the following option is **INCORRECT**?
(A) Dipole moment of H_2O_2 is greater than that of H_2O .
(B) O_2 is colourless in liquid as well as in gaseous phase.
(C) According to MOT, no $\sigma-$bond is present in C_2 molecule.
(D) On thermal decomposition of $Ca(NO_3)_2$, a paramagnetic gas is obtained as one of the product.

23. Which of the following organic compound is not soluble in aq. NaOH solution?
(A) 
(B) 
(C) 
(D) 

24. If 20 mL of 0.1 M $CH_3 - CH_2 - CH_2 - NH_2$ is titrated with 0.1 M HCl then which of the following indicator is most appropriate to find the equivalence point

$\left[pK_b\left(CH_3 - CH_2 - CH_2 - NH_2\right) = 3.3, \log 2 = 0.3\right]$

(A) methyl orange (pH range 3.2 – 4.4)
(B) methyl red (pH range 4.2 – 6.2)
(C) phenolphthalein (pH range 8.0 – 9.8)
(D) phenol red (pH range 3.2 – 4.4)

SECTION – A
(Matching List Type)

*This section contains **FOUR (04)** Matching List Type Questions. Each question has FOUR statements in **List-I** entries (P), (Q), (R) and (S) and FIVE statements in **List-II** entries (1), (2), (3), (4) and (5). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.*

25. Match the metal complexes ion in List-I with appropriate electronic configuration in List-II

List – I		List – II	
(P)	$\left[Cr(H_2O)_6\right]^{2+}$	(1)	$t_{2g}^4 e_g^o$
(Q)	$\left[Mn(en)_3\right]^{3+}$	(2)	$t_{2g}^3 e_g^1$
(R)	$\left[Fe(H_2O)_6\right]^{2+}$	(3)	$t_{2g}^4 e_g^2$
(S)	$\left[NiF_6\right]^{2-}$	(4)	$t_{2g}^5 e_g^1$
		(5)	$t_{2g}^6 e_g^o$

(A) P → 1; Q → 2; R → 3; S → 5
(B) P → 2; Q → 1; R → 3; S → 5
(C) P → 2; Q → 1; R → 5; S → 4
(D) P → 1; Q → 2; R → 4; S → 3

26. Match the following

List – I (Name of reaction)		List – II (Reaction)	
(P)	Reimer-Tiemann reaction	(1)	Phenol $\xrightarrow[\text{aq. NaOH}]{CHCl_3}$
(Q)	Stephen reaction	(2)	Alkyl nitrile $\xrightarrow[(ii)\ H_3O^+/\Delta]{(i)\ SnCl_2/HCl}$
(R)	Hunsdieker reaction	(3)	Acetophenone $\xrightarrow{NaOX/OH^-}$
(S)	Haloform reaction	(4)	Phenol $\xrightarrow[(iii)\ H^+]{(i)\ NaOH,\ (ii)\ CO_2}$
		(5)	Silver acetate $\xrightarrow[\Delta]{Br_2/CCl_4}$

(A) P → 4; Q → 5; R → 3; S → 1
(B) P → 1; Q → 5; R → 3; S → 2
(C) P → 4; Q → 1; R → 5; S → 3
(D) P → 1; Q → 2; R → 5; S → 3

27. Match the oxyacids in List-I to their anhydride in List-II

List – I (Oxyacids)		List – II (Acid anhydride)	
(P)	HOCl	(1)	Cl_2O
(Q)	HNO_3	(2)	N_2O_3
(R)	HNO_2	(3)	N_2O_5
(S)	$HClO_4$	(4)	Cl_2O_7
		(5)	N_2O

(A) P → 4; Q → 3; R → 2; S → 1
(B) P → 4; Q → 2; R → 5; S → 1
(C) P → 1; Q → 3; R → 2; S → 4
(D) P → 1; Q → 5; R → 3; S → 4

28. Match the following

List – I (Compound/Ion)		List – II (Shape)	
(P)	$[XeF_5]^-$	(1)	Square planar
(Q)	$[XeO_6]^{4-}$	(2)	Pentagonal planar
(R)	XeF_4	(3)	Octahedral
(S)	$[XeOF_5]^-$	(4)	Tetrahedral
		(5)	Pentagonal pyramidal

(A) P → 2; Q → 5; R → 4; S → 3
(B) P → 5; Q → 2; R → 1; S → 3
(C) P → 2; Q → 3; R → 1; S → 5
(D) P → 5; Q → 3; R → 4; S → 2

SECTION – B

(Numerical Answer Type)

*This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.*

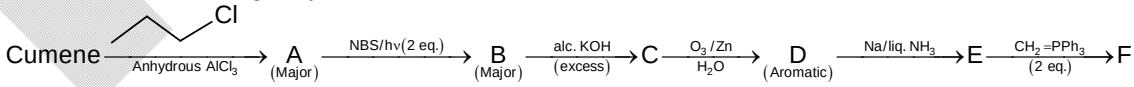
29. N_2 undergoes photochemical dissocation into excited nitrogen atom, each excited nitrogen atom has 2 eV higher energy than the unexcited N-atom. If bond dissociation energy of N_2 into unexcited 'N' atom is 960 kJ/mol and the maximum wavelength for photochemical dissociation is $x \times 10^{-11}$ m then the value of *x* is

Avogadro's number $= 6 \times 10^{23}$

$h = 6 \times 10^{-34}$ J – S

$C = 3 \times 10^{8}$ m / s

30. Consider the following sequence of reactions



Number of moles of CO_2 gas produced, when one mole of product 'F' is heated with O_3(excess) in presence of H_2O_2 is…….

31. If the mean life for $\alpha-$emission is $\frac{x}{3}$ years and mean life of $\beta-$emission of same radioactive substance is 400 years and time required for the decay of $7/8^{th}$ of radioactive sample is 500 years when the sample decaying α and β emission simultaneously. The value of x is:
$(\text{Given}: \ell n8 = 2)$.

32. How many of the following compound(s) is/are black in colour
PbS, CdS, CuS, CuI_2, Cu_2O, CuO, Sb_3S_2

33. The following data were obtained during kinetic study of the reaction in a closed vessel.
$2X(g) \longrightarrow 2Y(g) + 3Z(g) + 4W(s)$

Time (min)	0	50	100
Partial pressure of X (in mm Hg)	600	300	150

The total pressure (in mm Hg) after 200 min. is P_{total}. The value of $\frac{4}{5}P_{total}$ is

(Initially only X gas is present in the vessel)

34. How many of the following statement(s) is/are correct?
I. Disubstituted borazole $(B_3N_3H_4X_2)$ can have four isomer.
II. Orthoboric acid is used an antiseptic.
III. Oxidation state of $T\ell$ is +3 in $T\ell I_3$.
IV. Inorganic benzene is also known as borazon.
V. B_2H_6 is an example of nido borane.

Mathematics

PART – III

SECTION – A
(One or More than one correct type)

*This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

35. In ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ (a > b > 0) F is one of the foci and A, B are any two points out of four end points of its axes. If |FA| = 3, |FB| = 2, then all the possible values of |AB|
(A) 5 (B) 6
(C) $\sqrt{7}$ (D) $\sqrt{17}$

36. In a plane rectangular coordinate system, a circle and parabola $y^2 = 4x$ have just one common point and circle touches the x-axis at the focus F of parabola. The radius of circle and the point where circle touches parabola is/are
(A) radius = $\frac{4\sqrt{3}}{9}$ (B) point of contact $\left(\frac{1}{3}, \frac{2\sqrt{3}}{3}\right)$
(C) radius = $\frac{2\sqrt{3}}{9}$ (D) point of contact $\left(\frac{2}{3}, \frac{1}{3}\right)$

37. Given function $f(x)=\begin{cases}|\log_3 x-1| & ; \quad 0<x\le 9\\ 4-\sqrt{x} & ; \quad x>9\end{cases}$ defined on R^+, let a, b, c be three real numbers that are not equal to each other, satisfying f(a) = f(b) = f(c), (a < b < c), then
(A) ab = 9 (B) abc = 9c
(C) c $\in$ (9, 16) (D) abc $\in$ (81, 144)

SECTION – A
(One Options Correct Type)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

38. How many ordered pairs of positive real numbers (a, b) satisfy the equation
(1 + 2a)(2 + 2b)(2a + b) = 32ab
(A) 0 (B) 1
(C) 2 (D) an infinite number

39. Consider the set of complex numbers z satisfying $|1 + z + z^2| = 4$. The maximum value of the imaginary part of z is
(A) $\frac{\sqrt{17}}{2}$ (B) $\frac{\sqrt{21}}{2}$
(C) $\frac{\sqrt{13}}{2}$ (D) $\frac{\sqrt{19}}{2}$

40. Suppose 'a' is randomly selected from 1, 2, 3, ..., 10 and 'b' is randomly selected from –1, –2, –3, ... –10, then the probability of $a^2 + b$ divisible by 3 is
(A) $\frac{35}{100}$ (B) $\frac{37}{100}$
(C) $\frac{39}{100}$ (D) $\frac{38}{100}$

41. Suppose A, B are the vertices of the major axis of ellipse. E, F are the foci of ellipse. It is given that |AB| = 4, |AF| = $2+\sqrt{3}$ and P is a point on ellipse satisfying |PE|·|PF| = 2, then the area of ΔPEF is

(A) 4 (B) 3
(C) 2 (D) 1

SECTION – A
(Matching List Type)

*This section contains **FOUR (04)** Matching List Type Questions. Each question has FOUR statements in **List-I** entries (P), (Q), (R) and (S) and FIVE statements in **List-II** entries (1), (2), (3), (4) and (5). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.*

42. Match the following List-I with List-II

List – I		List – II	
(P)	$\vec{a}=2^m\hat{i}-\hat{j}$ and $\vec{b}=(2^m-1)\hat{i}+(2^m+1)\hat{j}$ is perpendicular where m is real number, then $\lvert\vec{a}\rvert$ is	(1)	$\frac{7}{25}$
(Q)	Given real number set {1, 2, 3, x} its greatest element is equal to the sum of its all elements, then the value of x is	(2)	0
(R)	Let $\alpha, \beta \in (0, \pi)$, cos α and cos β are the two roots of equation $5x^2 - 3x - 1 = 0$, then the value of sin α·sin β is	(3)	$\sqrt{4+2\sqrt{2}}$
(S)	Arranging five numbers (2, 0, 1, 9, 2019) on a line in any order to generate an 8 digit number (with the first digit no being zero), then the number of the different 8 digit numbers generated is	(4)	–3
		(5)	95

The correct option is:
(A) (P) $\rightarrow$ (3); (Q) $\rightarrow$ (4); (R) $\rightarrow$ (2); (S) $\rightarrow$ (5)
(B) (P) $\rightarrow$ (4); (Q) $\rightarrow$ (3); (R) $\rightarrow$ (2); (S) $\rightarrow$ (1)
(C) (P) $\rightarrow$ (4); (Q) $\rightarrow$ (3); (R) $\rightarrow$ (1); (S) $\rightarrow$ (5)
(D) (P) $\rightarrow$ (3); (Q) $\rightarrow$ (4); (R) $\rightarrow$ (1); (S) $\rightarrow$ (5)

43. Match the following List-I with List-II

List – I		List - II	
(P)	$\int_0^1 \frac{x^2 \ln x}{\sqrt{1-x^2}}dx$	(1)	$\frac{40\sqrt{2}}{\pi}$
(Q)	$\left\lvert\int_0^{10} \frac{\sin^{-1}([x]-x)}{\sqrt{1-([x]-x)^2}}dx\right\rvert$	(2)	$-\frac{3}{2}$
(R)	If f(x) = max{x – |x|, x + [x]}, then $\int_{-3}^{3} f(x)dx$ is	(3)	π
(S)	$\int_0^{20} \sqrt{1-\cos \pi x}$ is	(4)	$\frac{5\pi^2}{4}$
		(5)	$\frac{\pi}{8}(1-\ln 4)$

The correct option is:
(A) (P) → (2); (Q) → (4); (R) → (3); (S) → (1)
(B) (P) → (5); (Q) → (4); (R) → (2); (S) → (1)
(C) (P) → (5); (Q) → (4); (R) → (2); (S) → (3)
(D) (P) → (2); (Q) → (3); (R) → (4); (S) → (3)

44. Match the following List-I with List-II

List – I		List – II	
(P)	A function f is differentiable in [0, 5] such that f(0) = 4 and f(5) = –1, if $g(x)=\frac{f(x)}{x+1}$, then there exist some c ∈ (0, 5), then g′(c) is equal to	(1)	15
(Q)	Let LMVT is satisfied for $f(x)=\sqrt{25-x^2}$, and c ∈ (1, 5), then the value of c^2 is	(2)	0
(R)	Let f be differentiable ∀ x, if f(1) = –2, f′(x) ≥ 2 ∀ x ∈ [1, 6], then f(6) has least possible value is	(3)	2
(S)	Let f(x) and g(x) be differentiable for 0 ≤ x ≤ 1, f(0) = 2, g(0) = 0, f(1) = 6, let there exists a real number c ∈ (0, 1) such that f′(c) = 2g′(c), then g(1) is	(4)	8
		(5)	$-\frac{5}{6}$

The correct option is:
(A) (P) → (5); (Q) → (1); (R) → (3); (S) → (4)
(B) (P) → (4); (Q) → (3); (R) → (2); (S) → (1)
(C) (P) → (4); (Q) → (3); (R) → (1); (S) → (5)
(D) (P) → (5); (Q) → (1); (R) → (4); (S) → (3)

45. Match the following List-I with List-II

List – I		List – II	
(P)	The value of $^{50}C_{10}+{}^{49}C_{10}+{}^{48}C_{10}+.....+{}^{10}C_{10}$ is	(1)	$2^{100}-1$
(Q)	The value of $^{50}C_1+{}^{50}C_2+{}^{50}C_3+.....+{}^{50}C_{49}$ is	(2)	$^{53}C_{10}$
(R)	The value of $^{50}C_{10}+3\cdot{}^{50}C_9+3\cdot{}^{50}C_8+{}^{50}C_7$ is	(3)	$2^{50}-2$
(S)	The value of $^{50}C_{10}+4\cdot{}^{50}C_9+6\cdot{}^{50}C_8+4\cdot{}^{50}C_7+{}^{50}C_6$ is	(4)	$^{51}C_{11}$
		(5)	$^{54}C_{10}$

The correct option is:
(A) (P) → (2); (Q) → (4); (R) → (3); (S) → (1)
(B) (P) → (4); (Q) → (3); (R) → (2); (S) → (5)
(C) (P) → (4); (Q) → (3); (R) → (1); (S) → (5)
(D) (P) → (2); (Q) → (3); (R) → (4); (S) → (1)

SECTION – B

(Numerical Answer Type)

*This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.*

46. Let $\omega \neq 1$, be a 13th root of unity find the remainder when $\prod_{k=0}^{12}(2-2\omega^k+\omega^{2k})$ is divided by 1000

47. For a positive integer n, let $f_n(x)=\cos x\cos 2x\cos(3x).....\cos(nx)$. Find the smallest n such that $\left|f_n''(0)\right|>2023$

48. What is the product of all solution to the equation $\log_{7x}(2023)\cdot\log_{289x}(2023)=\log_{2023x}(2023)$

49. Suppose all the terms of arithmetic sequence $\{a_n\}$ are integers with the initial term $a_1 = 2019$ and for any positive integer n, there always exists positive integer m such that $a_1 + a_2 + ... + a_n = a_m$, then the number of such sequence is

50. In a plane rectangular co-ordinate system, suppose there exists a point (a, b) on the circle with centre (r + 1, 0) and radius r. Satisfying $b^2 \geq 4a$, then the minimum value of r is

51. For a given integer n > 4 in the expansion of $\left(x+2\sqrt{y}-1\right)^n$, the coefficients of terms x^{n-4} and xy are the same, then the value of n is